Computer Science Assignment

Topic: Computer Networks and IP Addressing

Submission Deadline: October 4, 2026, 11:59 PM

Answer all the following questions clearly. Include diagrams wherever required.

1. Explain the difference between IPv4 and IPv6. Give two advantages of IPv6.
2. What is an IP address? Explain the purpose of a subnet mask.
3. A computer has the IP address 192.168.10.25/24. Identify the network address, broadcast address, and usable host range.
4. Explain the difference between a router and a switch.
5. What happens when a packet is sent from one network to another through a router? Describe the basic steps.
6. Draw a simple network containing two PCs, one switch, and one router. Assign suitable IP addresses to the devices.

Name: _____

USN: _____

Section: _____